BIOLOGICAL OXYGEN DEMAND (BOD) 21

Biological (biochemical) oxygen demand is a very good indicator of the concentration of organic waste, bacteria and other aerobic organisms in a sample of water. The BOD of a sample of water can be determined in different ways. The following method, using the formula shown in figure 1, is a common one.

The **biological oxygen demand** (BOD) is the amount of dissolved oxygen consumed under specific conditions in a given time by the aerobic biological oxidation of organic (plant and animal) matter in a specific sample of water.

- Dilute the sample with distilled water (to ensure that not too much organic matter is present). Let the dilution factor = P.
- Measure the dissolved oxygen content. (D_1, in mg L^{-1})
- Incubate the diluted sample for 5 days at 20°C.
- Measure the dissolved oxygen content again. (D_2, in mg L^{-1})
- The BOD is then calculated by finding the difference between the dissolved oxygen contents and then dividing by the dilution factor, as shown in figure 1.

$$BOD = (D_1 - D_2) \div P$$

Figure 1

EXAMPLE

i **A sample of waste water was taken. 1 mL of this sample was made up to 100 mL using pure water. The D_1 of the diluted sample was found to be 7 mg L^{-1}. After 5 days incubation at 20°C, the D_2 was found to be 3 mg L^{-1}. Calculate the BOD for the sample.**

$BOD = (D_1 - D_2) \div P$

$= (7 - 3) \div 0.01$

$= 400$ mg L^{-1}

1 **A sample of waste water was taken. 1 mL of this sample was made up to 200 mL using pure water. The D_1 of the diluted sample was found to be 4 mg L^{-1}.** After 5 days incubation at 20°C, the dissolved oxygen content was found to be 2 mg L^{-1}. Calculate the BOD for the sample. Show your working.

Samples are incubated for five days because historically this was supposed to be the longest time that river water takes to flow from source to sea in the United Kingdom. The more oxygen that is used up, the greater the concentration of organic waste. If the BOD is high, decay organisms can use up all of the available oxygen, which harms other aquatic organisms.

2 In what ways can aquatic organisms be harmed if decay organisms use up the available oxygen?

3 *A measure of the mass of dissolved oxygen consumed in biological processes that degrade organic matter in surface water.* What does this define?

4 Figure 2 shows how the concentration of organic substances changes over time in the presence of organisms that consume organic material. Write an interpretation of what is shown in figure 2.

5 Explain why testing BOD of water samples from the output of a water treatment plant could indicate how efficient the plant is.

6 Write the missing word: *Organic matter that can be decomposed by biological means is said to be* _____ .

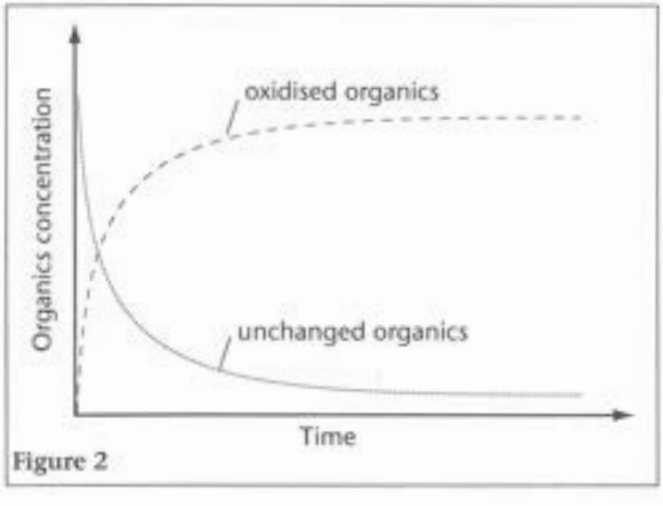


Figure 2

7 What is meant by *incubate*?

Research 8 Give your own definition of *biological oxygen demand*.

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